

Qatar-2b

R-0737

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_250205 · created 2026-07-28T01:31:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460711.9927 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.223 +/- 0.067
Transit depth (Rp/Rs)^2	4.95 +/- 2.98 [%]
Orbital Inclination (inc)	89.71 +/- 1.74
Airmass coefficient 1 (a1)	39.25 +/- 26.67
Airmass coefficient 2 (a2)	-3.0 +/- 0.54
Scatter in the residuals of the lightcurve fit is	87.04 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	4.0
Transit Duration (day)	0.0463 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.223 $\pm$ 0.067 vs 0.16327 (deviation 0.45 σ)
Duration fit vs published	0.0463 d vs 0.0767 d (deviation 5.34 σ)
Mid-time O-C	-4.0 min
Depth SNR	1.7
Residual scatter	87.04 %

Light curve

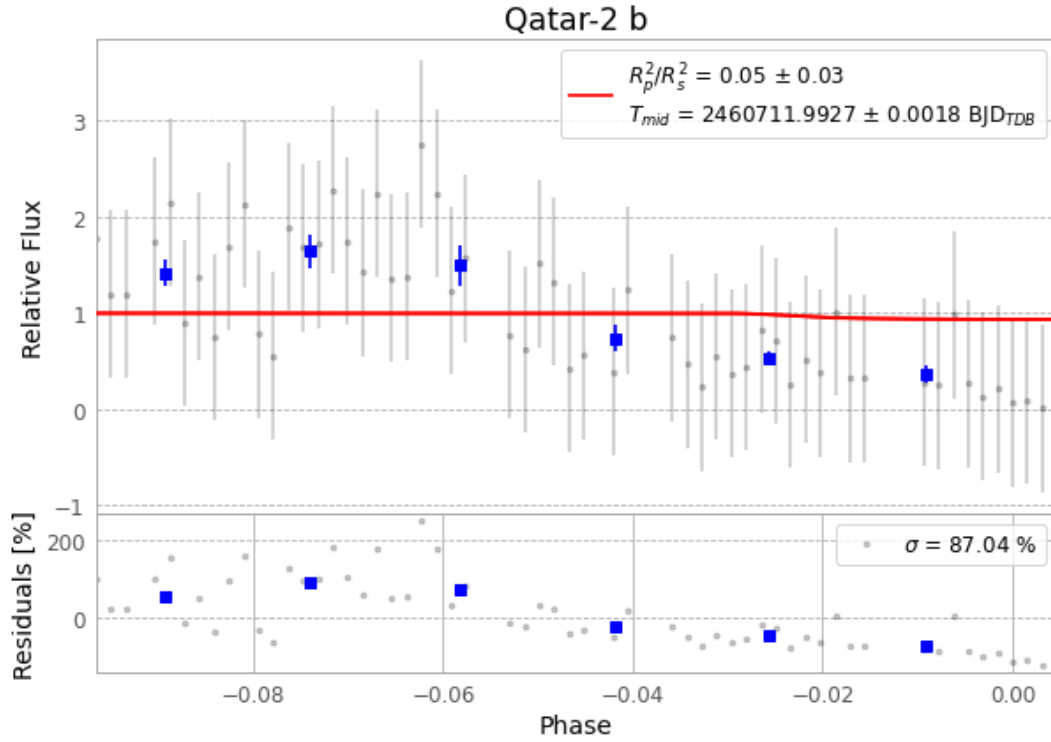


Figure 1 — FinalLightCurve_Qatar-2 b_2025-02-05.png (SHA-256 fcb584132d9ce205...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [497, 297], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 628958d8f97073c1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

65 FITS frames (43 MB), Qatar-2250205083918.FITS → Qatar-2250205115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-250205.log (cc290c679c29...), FinalLightCurve_Qatar-2 b_2025-02-05.png (fcb584132d9c...), FinalLightCurve_Qatar-2 b_2025-02-05.csv (625378a253cb...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 01:31Z.